

Module 2: Limits, Continuity & Derivatives

Ph.D. Math Camp

Introduction to Calculus

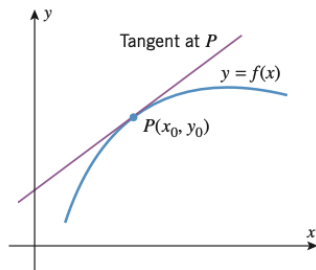
Limits

Limits give calculus its power, distinguishing it from algebra.

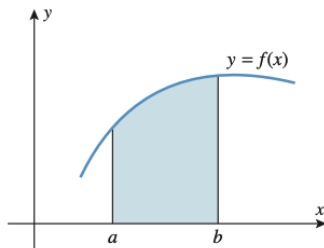
Many of the ideas of calculus originated with the following two geometric problems:

1. **The Tangent Line Problem:** Find the equation of the line tangent to a graph at a given point $P(x_0, y_0)$. → **Differential Calculus**
2. **The Area Problem:** Find the area between the graph of a function and an interval $[a, b]$ on the x-axis. → **Integral Calculus**

Geometric Problems



(a) Tangent Line



(b) Finding the Area

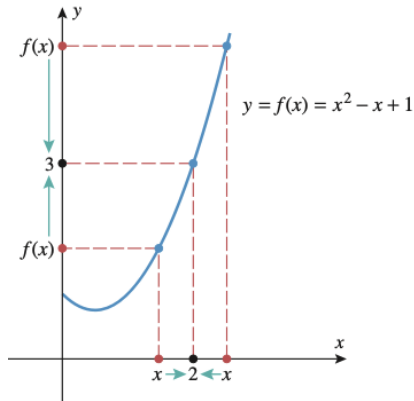
Understanding Limits

- ▶ Roughly speaking, finding the limit involves determining the value the function $f(x)$ is heading toward as x heads toward c .
- ▶ In this definition, x need not arrive at c , nor must $f(x)$ actually arrive at a destination as x heads toward c .
- ▶ If $f(x)$ is not heading anywhere, the limit does not exist. It's the direction that counts, not the destination.
- ▶ The most basic use of limits is to describe how a function behaves as the independent variable approaches a given value.

Example of Limit

Example: Examine the behavior of $f(x) = x^2 - x + 1$ as x approaches 2.

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x^2 - x + 1) = 3$$



Definition of a Limit

Formal Definition:

$$\lim_{x \rightarrow a} f(x) = L$$

- ▶ If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a
- ▶ Read “the limit of $f(x)$ as x approaches a is L ” or “ $f(x)$ approaches L as x approaches a .”

Example: Numerical Evidence

Example:

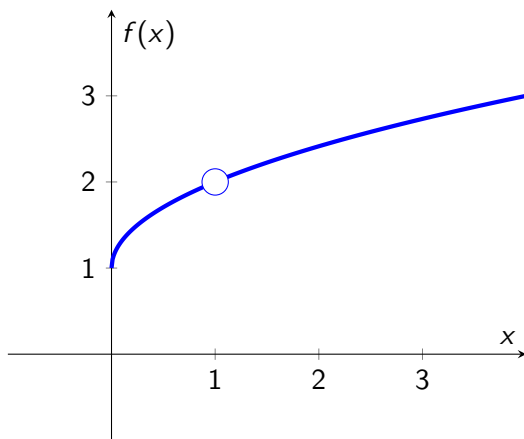
$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$$

Although the function is undefined at 1, this does not affect the limit.

Table: Input and Output Values as x Approaches 1

x	0.99	0.999	0.9999	0.99999	1.00001	1.0001	1.001	1.01
$f(x)$	1.994987	1.999500	1.999950	1.999995	2.000005	2.000050	2.000500	2.004988

Graphical Representation



The table and figure show $f(x)$ gets closer to 2, so

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = 2$$

Algebraic Properties of Limits

If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exist:

- ▶ $\lim_{x \rightarrow c} k = k$. Limit of a constant is the constant itself
- ▶ $\lim_{x \rightarrow c} [f(x) + g(x)] = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$
- ▶ $\lim_{x \rightarrow c} [f(x) - g(x)] = \lim_{x \rightarrow c} f(x) - \lim_{x \rightarrow c} g(x)$
- ▶ $\lim_{x \rightarrow c} [f(x)g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$
- ▶ $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, provided $\lim_{x \rightarrow c} g(x) \neq 0$
- ▶ $\lim_{x \rightarrow c} [f(x)]^p = [\lim_{x \rightarrow c} f(x)]^p$

Examples: Computing Limits

Compute the limits:

1. $\lim_{x \rightarrow 5} 9$

2. $\lim_{x \rightarrow 6} x^2$

3. $\lim_{x \rightarrow 3} 2x^3$

4. $\lim_{x \rightarrow 2} (x^4 + 3x)$

5. $\lim_{x \rightarrow 4} [(x + 8)(x - 5)]$

Examples: Computing Limits

Compute the limits:

1. $\lim_{x \rightarrow 5} 9 = 9$

2. $\lim_{x \rightarrow 6} x^2 = 36$

3. $\lim_{x \rightarrow 3} 2x^3 = 54$

4. $\lim_{x \rightarrow 2} (x^4 + 3x) = 22$

5. $\lim_{x \rightarrow 4} [(x + 8)(x - 5)] = -12$

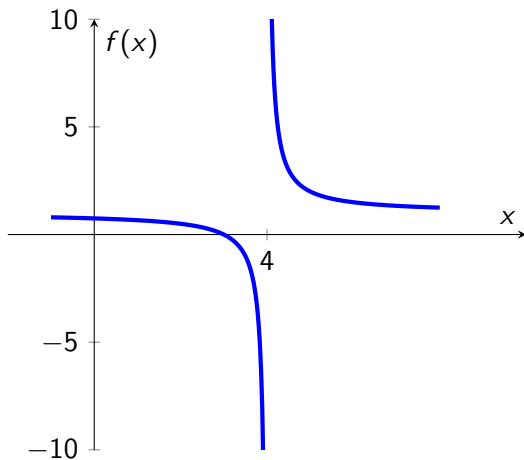
Limits of Polynomials and Rational Functions

If $p(x)$ and $q(x)$ are polynomials, then

- ▶ $\lim_{x \rightarrow c} p(x) = p(c)$
- ▶ $\lim_{x \rightarrow c} \frac{p(x)}{q(x)} = \frac{p(c)}{q(c)}$ and $q(c) \neq 0$
- ▶ If the denominator approaches zero and the numerator does not, the limit does not exist.

Example

- ▶ Consider $\lim_{x \rightarrow 4} \frac{x-3}{x-4}$.
- ▶ Plugging in $x = 4$ gives $\frac{1}{0}$.



Examples: Simplifying Limits

Example 1: $\lim_{x \rightarrow 7} \frac{x-7}{x^2-49}$

Example: Simplifying Limits

Example 1: $\lim_{x \rightarrow 7} \frac{x-7}{x^2-49}$

$$\lim_{x \rightarrow 7} \frac{x-7}{x^2-49} = \lim_{x \rightarrow 7} \frac{x-7}{(x-7)(x+7)}$$

$$= \lim_{x \rightarrow 7} \frac{\cancel{(x-7)}}{\cancel{(x-7)}(x+7)} = \lim_{x \rightarrow 7} \frac{1}{(x+7)} = \frac{1}{14}$$

Examples: Simplifying Limits

Example 2: $\lim_{x \rightarrow 6} \frac{x^2 - 2x - 24}{x - 6}$

$$\lim_{x \rightarrow 6} \frac{x^2 - 2x - 24}{x - 6} = \lim_{x \rightarrow 6} \frac{\cancel{(x - 6)}(x + 4)}{\cancel{(x - 6)}} = \lim_{x \rightarrow 6} (x + 4) = 10$$

One-Sided Limits

- ▶ Thus far, we have discussed **two-sided limits** because it requires the values of $f(x)$ to get closer and closer to L as values of x are taken from either side of $x = a$.
- ▶ However, some functions exhibit different behaviors on the two sides of an x -value a , in which case it is necessary to distinguish whether values of x near a are on the left side or on the right side of a for purposes of investigating limiting behavior.

One-Sided Limits

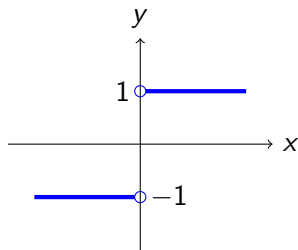
- ▶ **Two-Sided Limits:** Values of $f(x)$ get closer to L as x approaches a from both sides.
- ▶ **One-Sided Limits:** Values of $f(x)$ from either left or right of a .

$$\lim_{x \rightarrow c^-} f(x) = L, \quad \lim_{x \rightarrow c^+} f(x) = M$$

Example: One-Sided Limits

Consider the function:

$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x < 0 \end{cases}$$



Notice that the two-sided limit $\lim_{x \rightarrow 0} f(x)$ does not exist because the functional values $f(x)$ do not approach a single value L .

Evaluating One-Sided Limits

Evaluate:

$$\lim_{x \rightarrow 0^-} f(x), \quad \lim_{x \rightarrow 0^+} f(x)$$

- ▶ Two-sided limit $\lim_{x \rightarrow 0} f(x)$ does not exist as $f(x)$ does not approach a single value.
- ▶ The two-sided limit $\lim_{x \rightarrow a} f(x) = L$ exists if:

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

Limits Involving Infinity

If $f(x)$ approaches L as x increases or decreases without bound:

$$\lim_{x \rightarrow +\infty} f(x) = L, \quad \lim_{x \rightarrow -\infty} f(x) = M$$

- ▶ The behavior of a function $f(x)$ as x increases without bound or decreases without bound is sometimes called the end behavior of the function.

Unbounded Limits

Examples of unbounded growth or decline:

$$\lim_{x \rightarrow c^-} f(x) = +\infty, \quad \lim_{x \rightarrow c^+} f(x) = -\infty$$

- Sometimes, taking a limit results in unbounded growth or decline. Here we get situations like the following

Limits at Infinity

Limits involving powers:

► For $n > 0$:

► $\lim_{x \rightarrow +\infty} x^n = +\infty$

► $\lim_{x \rightarrow +\infty} -x^n = -\infty$

► For odd n :

► $\lim_{x \rightarrow -\infty} x^n = -\infty$

► For even n :

► $\lim_{x \rightarrow -\infty} x^n = +\infty$

► $\lim_{x \rightarrow \pm\infty} \frac{1}{x^n} = 0$

Polynomials at Infinity

The limit at infinity of a polynomial is determined by the highest power term:

$$\lim_{x \rightarrow +\infty} (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0) = \lim_{x \rightarrow +\infty} a_n x^n$$

Examples: Polynomials at Infinity

1. $\lim_{x \rightarrow +\infty} (3 - x^2 + 6x^4) = +\infty$

2. $\lim_{x \rightarrow +\infty} (3 - x^2 - 6x^4) = -\infty$

Limits at Infinity of Rational Functions

Some shortcuts to find the limit at infinity of a rational function

- ▶ If the degree of the numerator is equal to the degree of the denominator, the **limit equals the ratio of the leading coefficients**.
- ▶ If the degree of the numerator is less than the degree of the denominator, the limit is zero.
- ▶ If the degree of the numerator is greater than the degree of the denominator, the limit is infinity or negative infinity depending on the sign and power of the numerator

Examples: Finding Limits

Please find

1. $\lim_{x \rightarrow +\infty} \frac{3x^2 - 7x}{4x^2 - 21}$

2. $\lim_{x \rightarrow +\infty} \frac{4x^5}{x^3 + 2x}$

3. $\lim_{x \rightarrow +\infty} \frac{-x^3 + 2x + 1}{x - 3}$

Examples: Finding Limits

Please find

1. $\lim_{x \rightarrow +\infty} \frac{3x^2 - 7x}{4x^2 - 21}$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{3x^2}{x^2} - \frac{7x}{x^2}}{\frac{4x^2}{x^2} - \frac{21}{x^2}} = \lim_{x \rightarrow +\infty} \frac{3 - \frac{7}{x}}{4 - \frac{21}{x^2}} = \frac{3}{4}$$

2. $\lim_{x \rightarrow +\infty} \frac{4x^5}{x^3 + 2x} = +\infty$

3. $\lim_{x \rightarrow +\infty} \frac{-x^3 + 2x + 1}{x - 3} = -\infty$

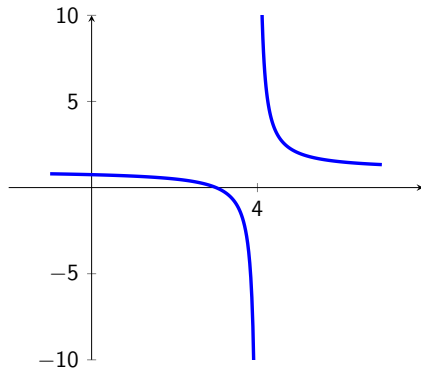
Introduction to Calculus (continued)

Asymptotes

- ▶ Limits involving infinity can describe graphical features called **asymptotes**.
- ▶ The graph of a function $f(x)$ has a **vertical asymptote** at $x = c$ if $f(x)$ increases or decreases without bound as x tends towards c , from either the right or the left.

Example: Vertical Asymptote

Recall, $\lim_{x \rightarrow 4} \frac{x-3}{x-4}$



As x approaches 4 from the left, $f(x)$ decreases without bound, but increases without bound if approaching from the right.

One-Sided Limit

The concept of one-sided limit summarizes this behavior:

$$\lim_{x \rightarrow 4^-} \frac{x-3}{x-4} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 4^+} \frac{x-3}{x-4} = +\infty$$

The line $x = c$ is a *vertical asymptote* of the graph of $f(x)$ if:

$$\lim_{x \rightarrow c^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow c^+} f(x) = \pm\infty$$

Vertical Asymptotes in Rational Functions

A rational function $R(x) = \frac{p(x)}{q(x)}$ has a vertical asymptote $x = c$ whenever $q(c) = 0$ but $p(c) \neq 0$.

Example: Vertical Asymptotes

Determine all vertical asymptotes:

1. $f(x) = \frac{x^2-9}{x^2+3x}$

- ▶ Set denominator to zero and solve: $x^2 + 3x = 0$
- ▶ Vertical asymptotes at $x = 0$. At $x = -3$ the numerator is 0.

2. $g(x) = \frac{3x^3-6x^2}{x^2-4}$

- ▶ Factor denominator: $(x+2)(x-2) = 0$
- ▶ Vertical asymptotes at $x = -2$. At $x = 2$ the numerator is 0.

Horizontal Asymptotes

When a function $f(x)$ tends towards a finite value b as x either increases or decreases without bound, the horizontal line $y = b$ is called the **horizontal asymptote** of the graph of $f(x)$.

Definition: Horizontal Asymptote

The horizontal line $y = b$ is called the **horizontal asymptote** of the graph if:

$$\lim_{x \rightarrow -\infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow +\infty} f(x) = b$$

Example: Vertical and Horizontal Asymptotes

Determine all vertical and horizontal asymptotes for

$$f(x) = \frac{3x^2}{x^2+2x-15}$$

► Horizontal asymptote: $\lim_{x \rightarrow +\infty} \frac{3x^2}{x^2+2x-15} = \lim_{x \rightarrow +\infty} \frac{3x^2}{x^2} = 3$

► Vertical asymptotes: Set denominator to zero
 $x^2 + 2x - 15 = 0 \implies (x+5)(x-3) = 0 \implies x = -5; x = 3$

Continuity

- ▶ Informally, a **continuous function** is one whose graph can be drawn without a pencil leaving the paper; that is, a graph with no holes or gaps.
- ▶ More formally, a function f is **continuous** at a point c if
 - ▶ $f(c)$ is defined
 - ▶ $\lim_{x \rightarrow c} f(x)$ exists
 - ▶ $\lim_{x \rightarrow c} f(x) = f(c)$

If each condition is met, then the function is continuous at c . If $f(x)$ is not continuous at c , it is said to have a **discontinuity** there. Polynomial or rational functions are continuous **whenever they are defined**.

Continuity

Examples: Functions with discontinuities

1. $f(x) = \frac{1}{x}$ at $x=0$
2. $f(x) = \frac{x-3}{x-4}$ at $x=4$

Continuity

Try these

1. Is $f(x) = 2x^2 + x - 2$ continuous at $x = 4$?
2. Is $f(x) = \frac{x-4}{x+5}$ continuous at $x = 2$?
3. Is $g(x) = \frac{x^2-4}{x-2}$ continuous at $x = 2$?
4. Is $g(x) = \frac{x^3-8}{x-2}$ continuous at $x = 2$?
5. Is $f(x) = \begin{cases} x^2 & \text{if } x \neq 2 \\ 5 & \text{if } x = 2 \end{cases}$ continuous at $x = 2$?

Continuity

1. Is $f(x) = 2x^2 + x - 2$ continuous at $x = 4$?

1.1 $f(x)$ defined? $f(4) = 34$ ✓

1.2 $\lim_{x \rightarrow 4} f(x) = 34$ ✓

1.3 $\lim_{x \rightarrow 4} f(x) = f(4)$ ✓

2. Is $f(x) = \frac{x-4}{x+5}$ continuous at $x = 2$?

2.1 $f(x)$ defined? $f(2) = \frac{-2}{7}$ ✓

2.2 $\lim_{x \rightarrow 2} f(x) = \frac{-2}{7}$ ✓

2.3 $\lim_{x \rightarrow 2} f(x) = f(2)$ ✓

Continuity

3. Is $g(x) = \frac{x^2-4}{x-2}$ continuous at $x = 2$?

3.1 $f(x)$ defined? $f(2) = \frac{0}{0}$ ✗

3.2 $\lim_{x \rightarrow 2} f(x) = 4$ ✓

3.3 $\lim_{x \rightarrow 2} f(x) \neq f(2)$ ✗

Continuity

4. Is $g(x) = \frac{x^3-8}{x-2}$ continuous at $x = 2$?

4.1 $f(x)$ defined? $f(2) = \frac{0}{0}$ ✗

4.2 $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x^2+2x+4)}{\cancel{(x-2)}} = 12$ ✓

4.3 $\lim_{x \rightarrow 2} f(x) \neq f(2)$ ✗

- To factor $x^3 - 8$, we can use the difference of cubes formula, which states that $a^3 - b^3$ can be factored as $(a - b)(a^2 + ab + b^2)$. In this case, $a = x$ and $b = 2$:

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$$

Continuity

5. $f(x) = \begin{cases} x^2 & \text{if } x \neq 2 \\ 5 & \text{if } x = 2 \end{cases}$

5.1 $f(x)$ defined? $f(2) = 5$ ✓

5.2 $\lim_{x \rightarrow 2} f(x) = 4$ ✓

5.3 $\lim_{x \rightarrow 2} f(x) \neq f(2)$ ✗

Continuity of a Function on an Interval

It is sometimes useful to have the following definitions of continuity on open and closed intervals.

1. A function f is said to be continuous on the open interval $a < x < b$ if it is continuous at each point x in the interval.
2. It is said to be continuous on the closed interval $a \leq x \leq b$ if f is continuous on the open interval $a < x < b$ and $f(x)$ approaches $f(a)$ as x approaches a from the right and $f(x)$ approaches $f(b)$ as x approaches b from the left.

In simpler terms— continuity on an interval means the graph of f is “one piece” throughout the interval.

Continuity of a Function on an Interval

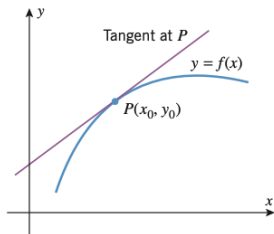
Example: Discuss the continuity of the function $f(x) = \frac{x+2}{x-3}$ on the open interval $-2 < x < 3$ and on the closed interval $-2 \leq x \leq 3$

- ▶ The function is continuous everywhere except 3. Thus, it is continuous within the open interval.
- ▶ However, $f(3)$ is undefined and the limit as it approaches 3 is infinite. Thus, not continuous on closed interval.

Derivatives

Derivatives

- ▶ Returning to the tangent problem. Suppose that we wanted an equation to the line that is tangent to $y = f(x)$ at point P



What would we need? Maybe we could use the point-slope formula.

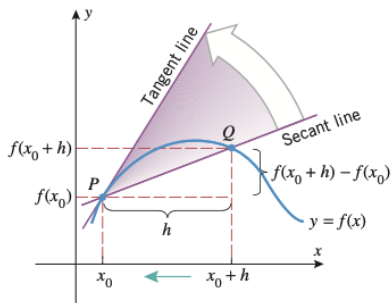
$$y - y_0 = m(x - x_0)$$

We could rewrite this as

$$y - f(x_0) = m(x - x_0)$$

Derivatives

What would we need? Maybe we could use the point-slope formula. $y - y_0 = m(x - x_0) \implies y - f(x_0) = m(x - x_0)$



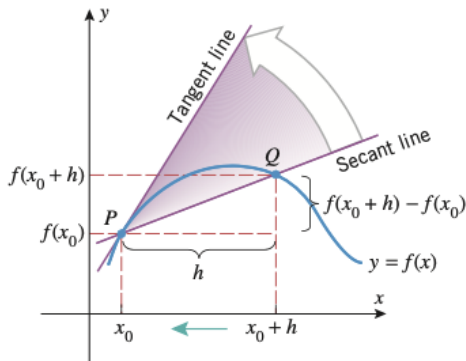
Derivatives

Here, we can use the two points on the secant line to get a poor approximation of the slope.

Point 1: $x_0, f(x_0) \Rightarrow (x_1, y_1)$

Point 2: $x_0 + h, f(x_0 + h) \Rightarrow (x_2, y_2)$

$$m_{sec} = \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0}$$

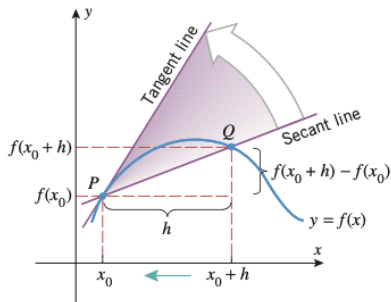


Derivatives

$$m_{sec} = \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0}$$

We can simplify the denominator

$$m_{sec} = \frac{f(x_0 + h) - f(x_0)}{h}$$

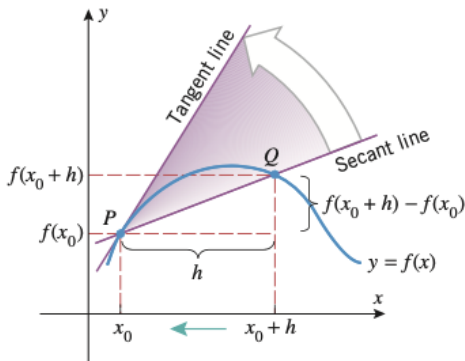


Derivatives

$$m_{\text{sec}} = \frac{f(x_0 + h) - f(x_0)}{h}$$

Suppose the true slope at point P is m_{tan} . It is clear that $m_{\text{sec}} < m_{\text{tan}}$ since the secant line is flatter.

- Could we do a better job with a second point closer to x_0 (i.e., smaller h)?

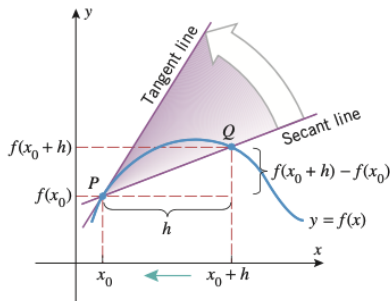


Derivatives

$$m_{sec} = \frac{f(x_0 + h) - f(x_0)}{h}$$

- ▶ Could we do a better job with a second point closer to x_0 (i.e., smaller h)? **Yes!** We could pick a smaller and smaller h until we get closer and closer to x_0 .
- ▶ In effect, to get m_{tan} we would need to take a limit as $h \rightarrow 0$.

Figure: Tangent + Secant



Derivatives

- ▶ Thus to get the slope m_{tan} at P we can take the limit of the secant equation as $h \rightarrow 0$

$$m_{tan} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

- ▶ Note that this limit may not exist. If it does, it is the slope of the tangent line at $(x_0, f(x_0))$.
- ▶ We call this the first derivative of f with respect to x and can denote it as $f'(x)$.

Derivatives

Formally, the limit definition of a derivative is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- ▶ When we want to find the slope or derivative of a non-linear function the slope will be changing as you move across the curve.
- ▶ Thus, typically, the derivative will result in a formula (another equation) that we can use to calculate the slope at any given point.

Derivatives

Use the limit definition of a derivative find the slope of the functions

1. $f(x) = x^2 + 2x + 7$ at $x = 2$.
2. $f(x) = \sqrt{x}$ at $x = 4$

Derivatives

1. $f(x) = x^2 + 2x + 7$ at $x = 2$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 2(x+h) + 7 - (x^2 + 2x + 7)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 2x + 2h + 7 - x^2 - 2x - 7}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 2h}{h}$$

$$= \lim_{h \rightarrow 0} (2x + h + 2)$$

$$= 2x + 2 \Big|_{x=2} = 6$$

Derivatives

2. $f(x) = \sqrt{x}$ at $x = 4$

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} &= \frac{\sqrt{x+h} - \sqrt{x}}{h} \left(\frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \right) \\ &= \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{4}} = \frac{1}{4}\end{aligned}$$

Derivatives

- ▶ We say that $f(x)$ is *differentiable* at c if $f'(c)$ exists i.e., the limit exists as $x \rightarrow c$.
- ▶ Another way to interpret the slope or f' is that the quantity f changes at the rate $f'(c)$ with respect to x when $x = c$. The derivative tells us the **rate of change** of the function with respect to x . We can also express the derivative of f as $\frac{df}{dx}$.

Techniques of Differentiation

The following are handy rules for finding a derivative and will be very useful in your coursework.

- ▶ **The constant rule:** For any constant c , $\frac{d}{dx}(c) = 0$. The derivative of a constant $f(x) = c$ is zero, it is a flat line with no slope.
- ▶ **The Power Rule:** For any real number n ,

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Reduce the exponent n of x by 1 and multiply by the original exponent.

- ▶ **The constant Multiple Rule:** If c is a constant and f is differentiable, then so is the function cf and

$$\frac{d}{dx}cf(x) = c\frac{df}{dx}$$

Techniques of Differentiation

- ▶ **The sum rule:** If $f(x)$ and $g(x)$ are differentiable, then so is the sum $S(x) = f(x) + g(x)$ and $S'(x) = f'(x) + g'(x)$; that is, the derivative of a sum is the sum of the derivatives.

$$\frac{d}{dx}[f(x) + g(x)] = \frac{df}{dx} + \frac{dg}{dx}$$

- ▶ **The product rule:** If $f(x)$ and $g(x)$ are differentiable at x , then so is their product $P(x) = f(x)g(x)$. Thus,

$$\frac{d}{dx}[f(x)g(x)] = \frac{df}{dx} \cdot g(x) + \frac{dg}{dx} \cdot f(x)$$

or equivalently.

$$(fg)' = gf' + fg'$$

In words, the derivative of the product is “the derivative of the first times the second plus the derivative of the second times the first.”

Techniques of Differentiation

- ▶ **The quotient rule:** If $f(x)$ and $g(x)$ are differentiable functions, then so is the quotient $Q(x) = \frac{f(x)}{g(x)}$

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{df}{dx} - f(x) \frac{dg}{dx}}{g^2(x)}$$

- ▶ In words, the derivative of the quotient is “the derivative of the top times the bottom minus the derivative of the bottom times the top all over the bottom squared.”
- ▶ Note that the quotient rule should not be used if you can easily use the power rule and product rule e.g.,

$$y = \frac{2}{x^3} - \frac{x}{3} - \frac{4}{5} + \frac{x+1}{x}$$

Techniques of Differentiation

Examples:

1. $y = 8x^3$
2. $f(x) = -6x^5$
3. $f(x) = 5x^2$
4. $y = \frac{7}{x}$
5. $f(x) = 18\sqrt{x}$
6. $f(x) = 5x^4(3x - 7)$
7. $f(x) = 7x^5 + x^3$
8. $f(x) = 5x^4 - (3x - 7)$
9. $f(x) = 7x^9(3x^2 - 12)$
10. $y = \frac{10x^8 - 6x^7}{2x}$
11. $y = \frac{3x^8 - 4x^7}{4x^3}$

Techniques of Differentiation

Chain rule: Suppose that $y = f(u)$ is a differentiable function of u , and $u = g(x)$ is a differentiable function of x . Then that means $y = f(g(x))$ is a composite function $f \circ g$ at x and

$$\frac{dy}{dx} = \frac{df}{du} \frac{du}{dx}$$

- ▶ That is, the derivative with respect to x is the derivative of f with respect to its argument u times the derivative of g with respect to its argument x .
- ▶ Practically, in functional form, we can write the chain rule as

$$\frac{d}{dx} (f(g(x))) = g'(x) f'[g(x)]$$

Techniques of Differentiation

$$\frac{dy}{dx} = \frac{df}{du} \frac{du}{dx}$$

- ▶ In words, the derivative of the inside times the derivative of the outside (applied to the inside).
- ▶ Another version of the chain rule applies to the following case (the general power rule)

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \frac{d}{dx}[g(x)]$$

Techniques of Differentiation

Find derivatives using chain rule

1. $f(x) = (3x^4 + 5)^6$

2. $y = \frac{1}{7x^3 + 13x^2 + 3}$

3. $y = \sqrt{34 - 6x^2}$

4. $y = \frac{(x+2)^2 \sqrt{x}}{2x}$

Revisit Tangent Line

1. Find the equation of the line tangent to the curve $y = x^3$ at the point where $x = -1$

► Find $\frac{dy}{dx} = 3x^2$, the slope formula.

► Find the point of the tangency $y(-1) = -1 \implies (-1, -1)$.

What is the slope at $x = -1$? $\frac{dy}{dx} = 3x^2 \Big|_{x=-1} = 3$

► Use point slope formula

$$(y - (-1)) = 3(x - (-1)) \implies y + 1 = 3x + 3$$

$$\implies \boxed{y = 3x + 2}$$

Revisit Tangent Line

Find the equation of the line tangent to the curve
 $y = (2x + 1)(x^2 - x + 3)$ at point $x = 0$.

Implicit Differentiation

- ▶ So far, the functions we have worked with have all been given by equations of the form $y = f(x)$ in which the dependent variable y on the left is given explicitly by an expression on the right involving the independent variable x . That is, the function is explicitly given on the right-hand side of the equation. A function in this form is said to be in explicit form.
- ▶ However, sometimes we will need to be able to take the derivative of a function of **implicit form**. In this form, the dependent variable has not been written explicitly in terms of the independent variable.

Implicit Differentiation

1. $y = \frac{x^{11}-x^4+3}{x^2} \rightarrow$ Explicit
2. $x^2y^5 = 2xy + y^3x \rightarrow$ Implicit

Implicit Differentiation

Suppose you have an equation that defines y implicitly as a function of x , and you want to find the derivative $\frac{dy}{dx}$.

- ▶ One possibility is that you may be able to solve the equation for y explicitly in terms of x . In such cases, we can determine the derivative explicitly.
- ▶ However, this may be impossible. Thus, we need a technique that allows us to solve for $\frac{dy}{dx}$ without having to solve for y explicitly.

The procedure for doing so is called **implicit differentiation**.

Implicit Differentiation

Suppose an equation defines y implicitly as a differentiable function of x . To find $\frac{dy}{dx}$

1. Differentiate both sides of the equation with respect to x , keeping in mind that y is a function of x that is not explicitly stated, and is a function of x . Thus, we must use the chain rule when differentiating terms containing $y = y(x)$.
2. Solve the differentiated equation algebraically for $\frac{dy}{dx}$. This expression is not explicit but involves x and y .

In implicit differentiation, you may find it helps to rewrite the original equation substituting " $y(x)$ " for " y " when differentiating.

Implicit Differentiation

Say we want to find $\frac{dy}{dx}$ for $3x^4 - 7y^5 = 86$.
Differentiate with respect to x to get

$$12x^3 \frac{dx}{dx} - 7(5y^4 \frac{dy}{dx}) = \frac{d}{dx} 86 \rightarrow 0$$

Simplifying we get,

$$12x^3 - 35y^4 \frac{dy}{dx} = 0 \implies 12x^3 = 35y^4 \frac{dy}{dx} \implies \boxed{\frac{dy}{dx} = \frac{12x^3}{35y^4}}$$

Implicit Differentiation

Find $\frac{dy}{dx}$

1. $x^4y^6 + 3y = 11,664$
2. $2x^3 + 5xy + 6y^2 = 13$
3. $(2x^3 + 7y)^2 = x^5$

Differentiation of Logarithmic and Exponential Functions

Exponential Functions

Let's find the derivative of $f(x) = e^x$ using the limit definition of a derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^{(x+h)} - e^x}{h}$$

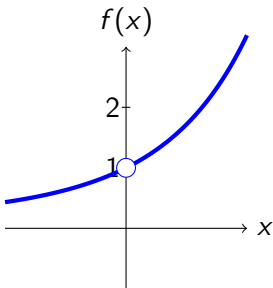
$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^x e^h - e^x}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h}$$

$$\Rightarrow e^x \cdot \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h}$$

$$e^x \cdot \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h}$$

- What is $\lim_{h \rightarrow 0} \frac{(e^h - 1)}{h}$? Let's try to find this graphically by taking an aside and rewriting the function as $\frac{(e^x - 1)}{x}$.



What is the limit as $x \rightarrow 0$? It is 1.

$$\Rightarrow \frac{d}{dx} e^x = e^x \cdot \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h} = e^x$$

$$\frac{d}{dx}e^x = e^x \cdot \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h} = e^x$$

Thus the derivative of e^x is itself. Geometrically this means that at each point $P(c, e^c)$ on the curve $y = e^x$, the slope is equal to e^c , the y coordinate of P .

The Chain Rule for Exponential Functions

If $u(x)$ is a differentiable function of x , then

$$\frac{d}{dx} \left[e^{u(x)} \right] = e^{u(x)} \frac{du}{dx}$$

Differentiate the following

1. $f(x) = e^{x^2}$
2. $f(x) = 3e^{7-2x}$
3. $y = 3xe^{2x}$

Logarithmic Functions

- ▶ Let's now find the derivative of the $\ln x$ function.
- ▶ Let $g(x) = \ln(x)$. Thus, if we exponentiate both sides we get $e^{g(x)} = x$

Let's take the derivative of both sides with respect to x (use chain rule on LHS)

$$g'(x)e^{g(x)} = 1. \text{ We know that } e^{g(x)} = x$$

$$g'(x)x = 1$$

$$\implies g'(x) = \frac{1}{x}$$

$$\implies g'(x) = \boxed{\frac{d}{dx} \ln(x) = \frac{1}{x}}, x > 0$$

Logarithmic Functions

Thus, $\frac{d}{dx} \ln(x) = \frac{1}{x}$ for $x > 0$.

Chain Rule for Logs

If $u(x)$ is a differentiable function of x , then

$$\frac{d}{dx}[\ln u(x)] = \frac{1}{u(x)} \frac{du}{dx} \text{ or, equivalently, } \frac{u'(x)}{u(x)}$$

Logarithmic Functions

Find derivatives

1. $f(x) = x \ln(x)$

2. $f(x) = \frac{\ln(\sqrt[3]{x^2})}{x^4}$

3. $y = 3xe^{2x}$

4. $f(x) = \ln 6x^2$

5. $f(x) = \ln(x^2 + 6x + 2)$

6. $y = \ln(1 + x)$

7. $y = 3 \ln(1 + x)^2$

Logarithmic Functions

Sometimes you can simplify differentiation if you first take a log.

$$g(x) = \frac{(3x^5-4)(2x^3+9)}{(7x^4-5)}$$

$$\ln(g(x)) = \ln((3x^5 - 4)(2x^3 + 9)) - \ln(7x^4 - 5)$$

$$\ln(g(x)) = \ln(3x^5 - 4) + \ln(2x^3 + 9) - \ln(7x^4 - 5)$$

$$\frac{d}{dx} \ln(g(x)) = \frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5}$$

$$\frac{g'(x)}{g(x)} = \frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5}$$

$$\Rightarrow g'(x) = \left(\frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5} \right) \cdot g(x)$$

$$g'(x) = \left(\frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5} \right) \left(\frac{(3x^5 - 4)(2x^3 + 9)}{(7x^4 - 5)} \right)$$

Logarithmic Functions

Simplify using logs and then find the derivative

1. $g(x) = \frac{(5x^3-8)(3x^4+7)}{(9x^5-2)}$

2. $g(x) = (x^3 - 2)(x^2 - 3)(8x - 5)$

3. $y = x^x$

L'Hôpital's Rule

L'Hôpital's Rule

- ▶ **L'Hôpital's Rule is useful when evaluating limits involving fractions where both the numerator and the denominator approach zero or infinity.**
- ▶ The rule allows us to find the limit of such expressions by taking the derivative of the numerator and denominator separately and then reevaluating the limit.

L'Hôpital's Rule

- Suppose we have a limit of the form $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$, where both $f(x)$ and $g(x)$ approach zero or infinity as x approaches c , and both functions are differentiable in some open interval containing c , except possibly at c itself.

L'Hôpital's Rule

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- ▶ If the limit of $\frac{f'(x)}{g'(x)}$ exists (or is infinite) as x approaches c , then the original limit is equal to the limit of $\frac{f'(x)}{g'(x)}$ as x approaches c .

L'Hôpital's Rule

- ▶ Suppose we have a limit of the form $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$, where both $f(x)$ and $g(x)$ approach zero or infinity as x approaches c , and both functions are differentiable in some open interval containing c , except possibly at c itself.
- ▶ If the limit of $\frac{f'(x)}{g'(x)}$ exists (or is infinite) as x approaches c , then the original limit is equal to the limit of $\frac{f'(x)}{g'(x)}$ as x approaches c .
- ▶ In other words, if the original limit is in an indeterminate form (e.g., $\frac{0}{0}$ or $\frac{\infty}{\infty}$), we can use L'Hôpital's Rule to evaluate.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

L'Hôpital's Rule

- ▶ For example, consider the limit $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$. If we try to substitute $x = 0$ directly, we get $\frac{0}{0}$, which is an indeterminate form.
- ▶ In such cases, L'Hôpital's Rule provides a way to find the limit by taking the derivatives of $e^x - 1$ and x separately and then reevaluating the limit. Applying L'Hôpital's Rule to this example, we find that the limit is equal to 1.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0} \frac{e^x}{1} = e^0 = 1$$

Evaluate the limit

1. $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$

2. $\lim_{t \rightarrow 1} \frac{5t^4 - 4t^2 - 1}{10 - t - 9t^3}$

3. $\lim_{x \rightarrow 0^+} x \ln x$

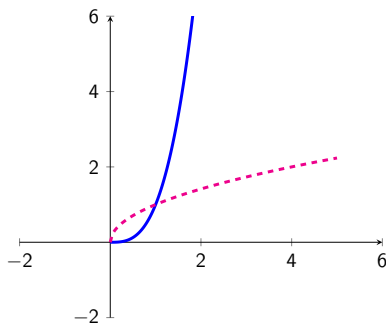
Second Derivatives

Second Derivatives

- ▶ The **second derivative** tells us the rate of change of the rate of change (or the rate of change in the slope). So, the second derivative of f at x is written as $f''(x)$ or $\frac{df'(x)}{dx}$ or $\frac{d^2f(x)}{dx^2}$.
- ▶ You get the second derivative by taking a derivative of the new function created that we called the first derivative.

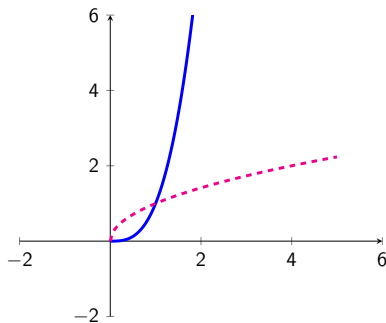
Second Derivatives

- ▶ This concept comes up repeatedly in public policy settings.
- ▶ For instance, demographers are interested in population growth and the change in the rate at which population growth (or decline) is taking place. Is it increasing at an increasing rate? Or is it increasing, but leveling off?



Second Derivatives

1. $f' > 0$, slope $\uparrow \implies f'' > 0$
2. $f' > 0$, slope $\downarrow \implies f'' < 0$
3. $f' < 0$, slope $\downarrow \implies f'' < 0$
4. $f' < 0$, slope $\uparrow \implies f'' > 0$



Second Derivatives

- ▶ $f''(x)$ is the derivative of $f'(x)$ read “ f double-prime x .”
- ▶ Note that there are no new rules to learn here. Just find the first derivative and then differentiate again.

Second Derivatives

1. Find the second derivative: $2x^3 + \frac{1}{x} + 7$
2. Find the second derivative: $y = (x^2 + 1)^5$
3. Suppose that a 5-year projection of population trends suggests that t years from now, the population of a certain community will be $P(t) = -t^3 + 9t^2 + 48t + 200$.
 - 3.1 At what rate will the population be growing 3 years from now?
 - 3.2 At what rate will the rate of population growth be changing with respect to time 3 years from now?

Second Derivatives

Suppose that a 5-year projection of population trends suggests that t years from now, the population of a certain community will be $P(t) = -t^3 + 9t^2 + 48t + 200$.

1. At what rate will the population be growing 3 years from now?

$$\left. \frac{dP(t)}{dt} = -3t^2 + 18t + 48 \right|_{t=3} = -3(3)^2 + 18(3) + 48 = 75$$

Second Derivatives

Suppose that a 5-year projection of population trends suggests that t years from now, the population of a certain community will be $P(t) = -t^3 + 9t^2 + 48t + 200$.

1. At what rate will the population be growing 3 years from now?

$$\frac{dP(t)}{dt} = -3t^2 + 18t + 48 \Big|_{t=3} = -3(3)^2 + 18(3) + 48 = 75$$

2. At what rate will the rate of population growth be changing with respect to time 3 years from now?

$$\frac{d^2P(t)}{dt^2} = -6t + 18 \Big|_{t=3} = 0$$

At $t = 3$, the growth rate is at its peak and begins to decrease after.

Higher-Order Derivatives

- ▶ In general, the derivative obtained from $f(x)$ after n successive differentiations is called the **n th derivative** or **derivative of order n** and is denoted by $f^{(n)}(x)$.
- ▶ For any positive integer n , the n th derivative of a function is obtained from the function by differentiating successively n times. If the original function is $y = f(x)$, the n th derivative is denoted by

$$\frac{d^n y}{dx^n} \text{ or } f^{(n)}(x)$$

Higher-Order Derivatives

Find the fourth derivative of the following

1. $f(x) = x^4 - x^3 + 1$

2. $f(x) = \frac{1}{x}$

3. $f(x) = xe^x$

Graphing

Graphing

- ▶ One way to visualize the behavior of a function of a single variable is to graph it.
- ▶ There are a number of methods for determining the characteristics of a graph.

Determining whether f is increasing or decreasing

- ▶ $f(x)$ is *increasing* at any point where $f'(x) > 0$
- ▶ $f(x)$ is *decreasing* at any point where $f'(x) < 0$
- ▶ $f(x)$ is *stationary* at any point where $f'(x) = 0$

Determining whether f is increasing or decreasing

- ▶ Procedure for finding intervals where f is increasing or decreasing
 1. Solve for $f'(x) = 0$ to determine the intervals of increase/decrease
 2. Plug in test numbers into $f'(x)$ on each interval to determine if the function is increasing or decreasing over that interval

Determining whether f is increasing or decreasing

Let's find the intervals of increase and decrease

1. $f(x) = 2x^3 + 3x^2 - 12x - 7$

$$f'(x) = 6x^2 + 6x - 12$$

f' is defined for all numbers. Set $f'(x) = 0$. These are going to be stationary points. **The function will change direction at these points.**

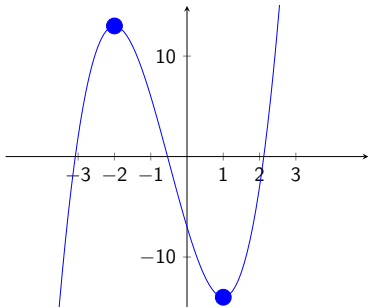
$$6x^2 + 6x - 12 = 0 \implies x^2 + x - 2 = (x - 1)(x + 2) = 0$$

$$\implies x = -2, x = 1$$

- Evaluate these points

$$f(-2) = 13 ; f(1) = -14$$

- This gives you the places where the curve will change direction.
- Plot these points. It is clear that they must be connected. The function at $(-2, 13)$ must fall for it to be a function analogous for $(1, -14)$.



Let's try $f(x) = \frac{x^2}{x-2}$

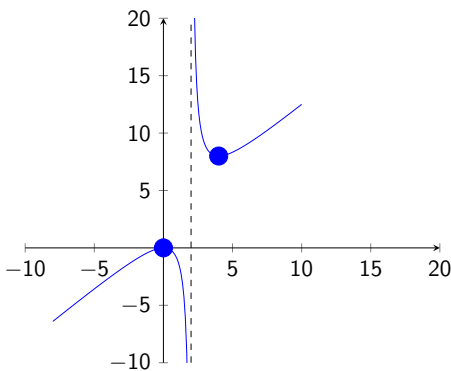
Let's try $f(x) = \frac{x^2}{x-2}$

$$f'(x) = \frac{2x(x-2) - x^2}{(x-2)^2} = \frac{x^2 - 4x}{(x-2)^2} = 0 \implies x = 0, x = 4$$

$$f(0) = 0, f(4) = 8$$

Try some points: $f(-1) = -\frac{1}{3}, f(1) = -1, f(3) = 9, f(5) = \frac{25}{3}$

Try some points: $f(-1) = -\frac{1}{3}$, $f(1) = -1$, $f(3) = 9$, $f(5) = \frac{25}{3}$

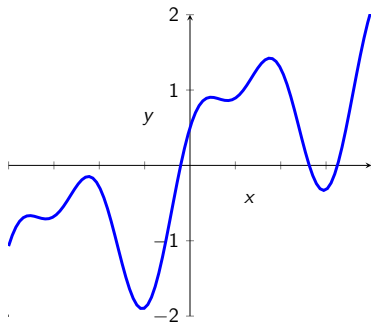


The function is **increasing** from $(-\infty, 0)$ and $(4, +\infty)$ and **decreasing** from $(0, 2)$ and $(2, 4)$

Relative Extrema

Relative Extrema

Let's consider a function with various peaks and valleys.



Notice that there are various peaks and valleys. Calculus can help us locate these points.

Relative Extrema

- ▶ A “peak” on the graph of a function is known as a **relative maximum**, and a “valley” is known as a **relative minimum**. Relative maxima and minima are collectively called **relative extrema**
- ▶ Formally, the graph of the function $f(x)$ is said to have a *relative maximum* at $x = c$ if $f(c) \geq f(x)$ for all x in an interval $a < x < b$ containing c . Similarly, the graph has a *relative minimum* at $x = c$ if $f(c) \leq f(x)$ on such an interval.

Relative Extrema

- ▶ But how do we know for which values a function has a relative extremum?
- ▶ If f has an extremum at x and f' exists there then $f'(x) = 0$. Of course, f' need not exist there.
- ▶ Points that satisfy either of these conditions are called **critical numbers**.

Relative Extrema

- Formally, a number c in the domain of $f(x)$ is called a **critical number** if either $f'(c) = 0$ or $f'(c)$ does not exist. The corresponding point $(c, f(c))$ on the graph of $f(x)$ is called a **critical point** for $f(x)$.

Relative Extrema

- ▶ This leads us to the **first derivative test for relative extrema**: Let c be a critical number for $f(x)$ and f be differentiable in an interval centered at c . Then the critical point $P(c, f(c))$ is:
 - ▶ A **relative maximum** if $f'(x) > 0$ to the left of c and $f'(x) < 0$ to the right of c
 - ▶ A **relative minimum** if $f'(x) < 0$ to the left of c and $f'(x) > 0$ to the right of c
 - ▶ **Not a relative extremum** if $f'(x)$ has the same sign on both sides of c

First-Derivative Test

Let's use the first derivative test to find all critical numbers for the function $f(x) = x^4 + 8x^3 + 16x^2 - 8$ and classify each as relative max, min, or neither.

First-Derivative Test

$$f'(x) = 4x^3 + 24x^2 + 32x \implies 4x(x^2 + 6x + 8) \implies 4x(x+2)(x+4) = 0$$

$$x = 0, x = -2, x = -4 \implies f(0) = -8, f(-2) = 8; f(-4) = -8$$

Thus, the critical points are

$$(x, y) = (0, -8), (-2, 8), (-4, -8)$$

x	-5	-4	-3	-2	-1	0	1
$f'(x)$	-60	0	12	0	-12	0	60

First-Derivative Test

Thus, the critical points are

$$(x, y) = (0, -8), (-2, 8), (-4, -8)$$

x	-5	-4(min)	-3	-2 (max)	-1	0 (min)	1
$f'(x)$	-60	0	12	0	-12	0	60

The first derivative test will always give us the nature of the relative extrema. However, there is a shortcut that works much of the time. This is the **second derivative test**.

Second Derivative Test

Suppose that $f'(a) = 0$ and f'' exists and is continuous in an interval centered around a , then:

- ▶ Suppose $f''(a) > 0$, then f has a relative minimum at $x = a$
- ▶ If $f''(a) < 0$, then f has a relative maximum at $x = a$

However, if $f''(a) = 0$ or if $f''(a)$ does not exist, the test is inconclusive.

- ▶ Limitation of the second derivative test
 - ▶ Sometimes computing the second derivative is messy
 - ▶ The test only applies to critical points at which the first derivative is zero, and not to those where the derivative is undefined.
 - ▶ If both $f'(c) = 0$ and $f''(c) = 0$, the second derivative test is inconclusive.
- ▶ In these cases, we can always use the first derivative test to find the nature of the critical point.
- ▶ If both $f'(c) = 0$ and $f''(c) = 0$, then we can use another shortcut– the **successive derivative test**– to learn the nature of $(c, f(c))$.

Successive derivative test

If both $f'(c) = 0$ and $f''(c) = 0$, then $(c, f(c))$ is:

- ▶ An **inflection point** if the first nonzero value of a higher-order derivative is an **odd-numbered derivative**. An inflection point of a continuous function is a point where the function changes from “cap-like” to “cup-like” or vice-versa.
- ▶ A **relative maximum** if the first nonzero value of a higher-order derivative is an **even-numbered** derivative with a negative value.
- ▶ A **relative minimum** if the first nonzero value of a higher-order derivative is an **even-numbered** derivative with a positive value.

Find relative extrema and identify as min or max for the following functions

1. $f(x) = -7x^2 + 126x - 23$

2. $f(x) = x^3 - 12x^2 + 36x + 8$

3. $y = (5 - x)^3$

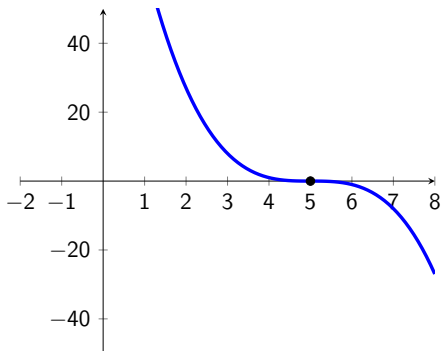
For #3: $y = (5 - x)^3$,

$$\frac{dy}{dx} = 3(-1)(5 - x)^2 \implies x = 5$$

Check $x = 5$ in $y'' \implies y''(x) = -6(-1)(5 - x) \Big|_{x=5} = 0$

- ▶ This means the second-derivative test is inconclusive. We can try the first derivative test. However, $y'(4) < 0$ and $y'(6) < 0$.
- ▶ If we take another derivative, we get $y''' = -6$. This is the first nonzero value of a higher-order derivative, and it is an **odd-numbered derivative (3)**. Thus, this is an **inflection point**.

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- Note that this is neither a relative maximum nor minimum.
- We get a saddle point or a critical point of a function where the function's derivative is zero, but it is neither a relative minimum nor a relative maximum.

- ▶ We get a saddle point or a critical point of a function where the function's derivative is zero, but it is neither a relative minimum nor a relative maximum.
- ▶ At a saddle point, the function changes concavity, and the nearby points on one side of the saddle point are higher, while the nearby points on the other side are lower.

Try these

1. $y = -(x - 8)^4$
2. $y = -2(x - 6)^6$
3. $y = x^2 e^x$
4. The total cost of producing Q units of a commodity is

$$C(Q) = 2Q^2 + 10Q + 32$$

Find the value of Q which minimizes average total cost.

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- . Find the value of Q which minimizes average total cost.

$$A(Q) = C(Q)/Q = 2Q + 10 + \frac{32}{Q}$$

$$A'(Q) = 2 - \frac{32}{Q^2} = 0 \implies 2Q^2 = 32 \implies Q^* = 4$$

$$A''(Q) = \frac{64}{Q^3} \bigg|_{Q=4} > 0$$

$\implies$ minimum

Concavity and Convexity

Concavity and Convexity

- ▶ The following notions of **concavity** and **convexity** describe the increase and decrease of the slope of the tangent to a curve.
- ▶ If the function $f(x)$ is differentiable on the interval $a < x < b$, then the graph of f is:
 - ▶ **Convex** on $a < x < b$ if $f'(x)$ is increasing on the interval (“cup-like” or hold water)
 - ▶ **Concave** on $a < x < b$ if $f'(x)$ is decreasing on the interval (“cap-like” or spills water)

Concavity and Convexity

There is a simple characterization of concavity/convexity in terms of the sign of the second derivative.

Test for Concavity

- ▶ If $f''(x) > 0$ on the interval $a < x < b$, then f is convex on this interval
- ▶ If $f''(x) < 0$ on the interval $a < x < b$, then f is concave on this interval

We should not confuse concavity/convexity of a curve with its increase or decrease! A curve that is convex on an interval may be either increasing or decreasing on that interval. The same goes for concavity.

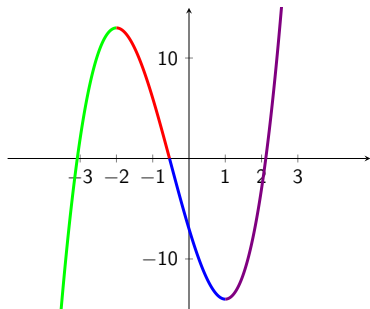
Concavity and Convexity

► $\uparrow$ **Concave**

► $\downarrow$ Concave

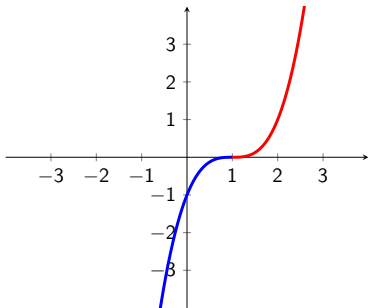
► $\downarrow$ **Convex**

► $\uparrow$ Convex



Determining Concavity and Locating Inflection Points

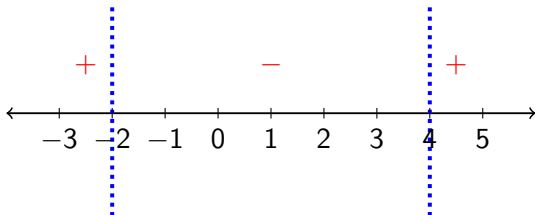
- ▶ Find f'' and determine numbers where $f''(x) = 0$ or $f''(x)$ does not exist
- ▶ Mark the numbers on a number line, dividing the line into sub-intervals. Evaluate $f''(p)$ for a test number p in each subinterval.
- ▶ The graph is convex on those intervals where $f''(p) > 0$ and concave where $f''(p) < 0$. The inflection points are the subdivision points where the concavity changes.



- ▶ Determine where the function $f(x) = x^3 - 3x^2 - 24x$ is increasing and decreasing, and where its graph is concave and convex.
- ▶ Calculate the first two derivatives:

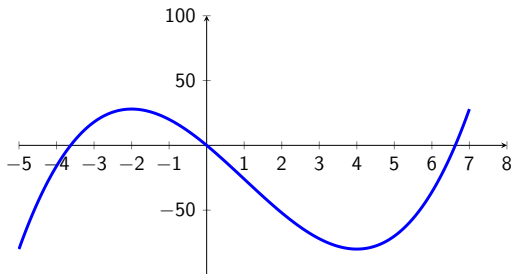
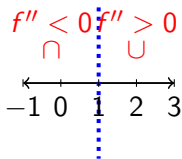
$$f'(x) = 3x^2 - 6x - 24 = 0$$

$$\implies 3(x^2 - 2x - 8) = 3(x - 4)(x + 2) = 0 \implies x = 4, x = -2$$



Second $\rightarrow$

$$f''(x) = 6x - 6 = 0 \implies x = 1$$



- ▶ Recall that a point on the graph of a function $f(x)$ where the concavity changes is called an **inflection point**.
- ▶ At an inflection point, either $f''(c) = 0$ or $f''(c)$ does not exist. The sign of the first derivative is immaterial.
- ▶ However, just because $f''(c) = 0$, you cannot necessarily conclude that $(c, f(c))$ is an inflection point.
- ▶ For example, $f(x) = x^4$, then $f''(x) = 12x^2$, and the graph of f is always concave up, even though $f''(0) = 0$.

Concavity and Convexity

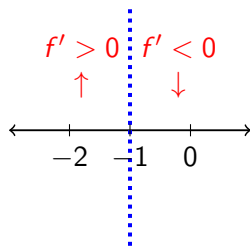
Let's try this one: $y = (x + 2)e^{-x}$

Concavity and Convexity

$$y = (x + 2)e^{-x}$$

$$\begin{aligned}y' &= e^{-x} - (x + 2)e^{-x} = 0 \\&= e^{-x}(1 - x - 2) = 0 = e^{-x}(-1 - x) \\&= (-1 - x) = 0 \implies x = -1 \\y''(-1) &< 0\end{aligned}$$

Increase/Decrease

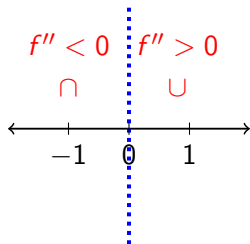


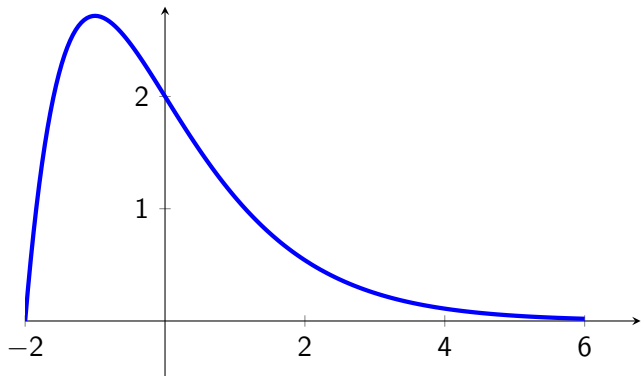
Second derivative

$$y'' = -e^{-x}(-1 - x) - e^{-x} = 0$$

$$x + 1 - 1 = 0 \implies x = 0$$

Concave/convex





General Procedure for Sketching a Graph

1. Find the domain of $f(x)$ (that is, where $f(x)$ is defined)
2. Find and plot all intercepts. The y -intercept (where $x = 0$) is usually easy to find, but the x -intercepts (where $f(x) = 0$).
3. Determine all vertical and horizontal asymptotes of the graph. Draw the asymptotes in a coordinate plane
4. Find $f'(x)$ and use it to determine the critical number of $f(x)$ and intervals of increase and decrease
5. Determine all relative extrema (both coordinates). Plot each relative maximum with a “cap” and each relative minimum with a “cup.”
6. Find $f''(x)$ and use it to determine intervals of concavity/convexity and inflection points. Plot each inflection point with a “twist” to suggest the shape of the graph near the point.
7. Connect points/lines on the plot.

Draw graph

Draw the graph of $f(x) = \frac{x}{(x+1)^2}$

Draw graph

Draw the graph $f(x) = \frac{x}{(x+1)^2}$

1. Domain all $x \neq -1 \implies$ discontinuity
2. Plot intercepts.

$$\frac{x}{(x+1)^2} = 0 \implies x = 0 \text{ For x-intercept}$$

$$\frac{0}{(0+1)^2} = y \implies y = 0 \text{ For y-intercept}$$

3. Asymptotes

For vertical asymptotes, we see that

$$\lim_{x \rightarrow -1} \frac{x}{(x+1)^2} = -\infty$$

For horizontal asymptotes,

$$\lim_{x \rightarrow \pm\infty} \frac{x}{(x+1)^2} = 0$$

4. Find critical numbers and intervals of increase and decrease

$$f'(x) = \frac{(x+1)^2 - 2x(x+1)}{(x+1)^4} = 0$$

$$\implies x^2 + 2x + 1 - 2x^2 - 2x = 0 \implies -x^2 + 1 = 0 \implies x = \pm 1$$

$$\implies x = 1, f(1) = \frac{1}{4}$$

$$f'(.5) > 0, f'(2) < 0 \implies \uparrow \in (-1, 1) \quad \downarrow \in (1, \infty)$$

5. Relative max at $(1, \frac{1}{4})$

6. Find $f''(x)$ to determine concavity and inflection points

$$f''(x) = \frac{-2x(x+1)^4 - 4(x+1)^3(-x^2+1)}{(x+1)^8} = 0$$

$$\implies (x+1)^3(-2x^2 - 2x + 4x^2 - 4) = 0$$

$$\implies 2(x+1)^3(x^2 - x - 2) = 0 \implies x = 2$$

Determine concavity $f''(1) < 0, f''(3) > 0$. The graph is concave on $(-\infty, -1)$ and on $(-1, 2)$ (recall $x = -1$ is not in the domain) and convex on $(2, \infty)$

7. Plot figure

